

Install and configure Apache, Nginx, PHP-31

Introduction

Apache and Nginx are popular web servers used in Linux. PHP is a server-side scripting language used for dynamic websites. In this task, we will install Apache, Nginx, and PHP on RHEL, configure them, and test with sample files.

Install Apache HTTP Server

```
sudo yum install httpd -y
```

Start and Enable

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

Verify Service

```
systemctl status httpd
```

Test

- Place an index file:

```
sudo vi /var/www/html/batch8.html
```

```
or touch /var/www/html/batch8.html
```

- Open browser: `http://<server-ip>`

3. Install Nginx

Install

```
sudo yum install nginx -y
```

Start and Enable

```
sudo systemctl start nginx
```

```
sudo systemctl enable nginx
```

Verify Service

```
systemctl status nginx
```

Test

```
sudo nano /usr/share/nginx/html/batch8.html
```

- Open browser: <http://192.168.163.134/batch8.html>

These are the commands used in my machine

```
170 sudo yum install httpd -y
171 sudo systemctl status httpd
172 sudo systemctl start httpd
173 sudo systemctl enable httpd
174 sudo systemctl status httpd
175 sudo firewall-cmd --permanent --add-service=http
176 sudo firewall-cmd --permanent --add-service=https
177 sudo firewall-cmd --reload
178 sudo systemctl status httpd I
```

```
[bharath9399@localhost ~]$ sudo systemctl start httpd
[bharath9399@localhost ~]$ sudo systemctl status httpd
• httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Thu 2025-08-21 11:15:37 IST; 11s ago
     Docs: man:httpd.service(8)
  Main PID: 38738 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 48533)
   Memory: 13.9M
      CPU: 99ms
   CGroup: /system.slice/httpd.service
           └─38738 /usr/sbin/httpd -DFOREGROUND
             └─38739 /usr/sbin/httpd -DFOREGROUND
               └─38740 /usr/sbin/httpd -DFOREGROUND
                 └─38741 /usr/sbin/httpd -DFOREGROUND
                   └─38742 /usr/sbin/httpd -DFOREGROUND

Aug 21 11:15:37 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Aug 21 11:15:37 localhost.localdomain httpd[38738]: AH00558: httpd: Could not reliably determine the s
Aug 21 11:15:37 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Aug 21 11:15:37 localhost.localdomain httpd[38738]: Server configured, listening on: port 80
```

```
[bharath9399@localhost ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/htt
pd.service.
[bharath9399@localhost ~]$ sudo firewall -cmd --permanent --add-service=http
sudo: firewall: command not found
[bharath9399@localhost ~]$ sudo firewall-cmd --permanent --add-service=http
success
[bharath9399@localhost ~]$ sudo firewall-cmd --permanent --add-service=https
success
[bharath9399@localhost ~]$ sudo firewall-cmd --reload
success
[bharath9399@localhost ~]$ cd /var/www/html
[bharath9399@localhost html]$ ls
[bharath9399@localhost html]$ touch Batch8.html
touch: cannot touch 'Batch8.html': Permission denied
[bharath9399@localhost html]$ sudo su
[root@localhost html]# ls
[root@localhost html]# cd /var/www/html
[root@localhost html]# ls
[root@localhost html]# touch batch8.html
[root@localhost html]# ls
batch8.html
[root@localhost html]# nano batch8.html
[root@localhost html]#
```

```
36 cd /var/www/html
37 ls
38 touch batch8.html
39 ls
40 nano batch8.html
41 history I
42 exit
43
```

```
19 <html>
20 <head>
21 <title>Batch8 Testing Apache</title>
22 </head>
23 <body>
24 <h1> This is the testing server for apache-Batch8</h1>
25 </body>
26 </html>
```



Installing Nginx

```
sudo yum install nginx -y
```

Start and Enable

```
sudo systemctl start nginx
```

```
sudo systemctl enable nginx
```

Verify Service

`systemctl status nginx`

Test

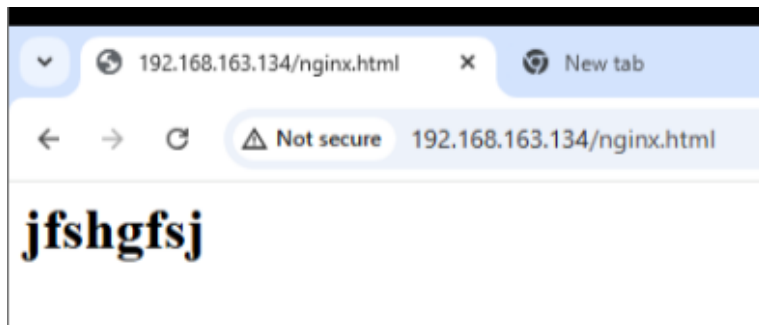
`sudo vim /usr/share/nginx/html/nginx.html`

- Open browser: <http://192.168.163.134/nginx.html>

```
64 sudo yum install epel-release -y
65 sudo yum install nginx -y
66 sudo systemctl status nginx
67 sudo systemctl start nginx
68 sudo systemctl stop httpd
69 sudo systemctl start nginx
70 sudo systemctl status nginx
71 sudo firewall-cmd --permanent --add-service=http
72 sudo firewall-cmd --permanent --add-service=https
73 sudo firewall-cmd --reload
74 sudo systemctl status httpd
75 sudo systemctl status httpd
76 sudo systemctl status nginx
77 sudo systemctl status nginx
78 sudo systemctl status nginx
79 cd /usr/share/nginx/html/
80 touch nginx.html
```

```
Last login: Wed Aug 20 09:32:09 2025 from 192.168.163.1
[bharath9399@localhost ~]$ sudo su
[sudo] password for bharath9399:
[root@localhost bharath9399]# sudo systemctl start nginx
[root@localhost bharath9399]# sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Drop-In: /etc/systemd/system/nginx.service.d
            └─php-fpm.conf
   Active: active (running) since Thu 2025-08-21 12:03:10 IST; 3min 23s ago
     Process: 41132 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 41133 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 41134 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 41135 (nginx)
      Tasks: 2 (limit: 48533)
     Memory: 2.2M
        CPU: 44ms
    CGroup: /system.slice/nginx.service
            └─41135 "nginx: master process /usr/sbin/nginx"
               └─41136 "nginx: worker process"

Aug 21 12:03:10 localhost.localdomain systemd[1]: Starting The nginx HTTP and reverse proxy server...
Aug 21 12:03:10 localhost.localdomain nginx[41133]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 21 12:03:10 localhost.localdomain nginx[41133]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 21 12:03:10 localhost.localdomain systemd[1]: Started The nginx HTTP and reverse proxy server.
[root@localhost bharath9399]# sudo firewall-cmd --permanent --add-service=http
Warning: ALREADY_ENABLED: http
success
[root@localhost bharath9399]# sudo firewall-cmd --permanent --add-service=https
Warning: ALREADY_ENABLED: https
success
[root@localhost bharath9399]# sudo firewall-cmd --reload
success
[root@localhost bharath9399]# cd /usr/share/nginx/html/
[root@localhost html]# ls
404.html  50x.html  icons     index.html  nginx-logo.png  poweredby.png  system_noindex_logo.png
[root@localhost html]# touch nginx.html
[root@localhost html]# nano index.html
[root@localhost html]# touch nginx.html
[root@localhost html]# ls
404.html  50x.html  icons     index.html  nginx.html  nginx-logo.png  poweredby.png  system_noindex_logo.png
[root@localhost html]# nano nginx.html
[root@localhost html]#
```



PHP installation & configuration:

Enable EPEL and Remi repositories

```
sudo yum install epel-release yum-utils -y
```

```
sudo yum install https://rpms.remirepo.net/enterprise/remi-release-7.rpm -y
```

- EPEL (Extra Packages for Enterprise Linux) provides additional packages not included in default RHEL repos.
- Remi repo provides updated PHP versions.

Configure Yum/DNF

```
sudo yum-config-manager --disable 'remi-php*'
```

```
sudo dnf install epel-release -y
```

```
sudo dnf install https://rpms.remirepo.net/enterprise/remi-release-9.rpm -y
```

Disables all PHP streams first, then adds the correct repo for PHP.

Enable specific PHP version

```
sudo dnf module reset php
```

```
sudo dnf module enable php:remi-8.3 -y
```

- Resets any previously installed PHP modules.
- Enables PHP 8.3 from Remi.

Install PHP and required extensions

```
sudo dnf install php php-cli php-fpm php-mysqlnd php-xml php-mbstring php-json php-zip -y
```

- Installs PHP core + commonly used modules:
 - php-cli → Command-line PHP
 - php-fpm → FastCGI Process Manager for performance
 - php-mysqlnd → MySQL support
 - php-xml → XML handling
 - php-mbstring → Multi-byte string handling
 - php-json → JSON support
 - php-zip → ZIP support

Check PHP version

```
php -v
```

Edit PHP configuration

```
sudo vim /etc/php.ini
```

Default PHP settings (upload size, timezone, etc.) can be modified here.

Enable PHP-FPM service

```
sudo systemctl enable --now httpd && sudo systemctl enable --now php-fpm
```

Starts Apache (httpd) and PHP-FPM.

Create test PHP file

```
sudo vim /var/www/html/info.php
```

Typically, the file contains:

```
<?php
phpinfo();
?>
```

Used to verify PHP is working.

Open firewall for web traffic

```
sudo firewall-cmd --permanent --add-port=80/tcp --add-port=443/tcp
```

```
sudo firewall-cmd --reload
```

Opens HTTP (80) and HTTPS (443) ports.

Open: <http://192.168.163.134/info.php>

```
46 sudo yum install epel-release yum-utils -y
47 sudo yum install https://rpms.remirepo.net/enterprise/remi-release-7.rpm -y
48 sudo yum install -y https://rpms.remirepo.net/enterprise/remi-release-7.rpm
49 sudo dnf install -y https://rpms.remirepo.net/enterprise/remi-release-8.rpm
50 sudo yum -y install yum-utils
51 sudo yum-config-manager --disable 'remi-php*'
52 sudo dnf install epel-release -y
53 sudo dnf install https://rpms.remirepo.net/enterprise/remi-release-9.rpm -y
54 sudo dnf module reset php
55 sudo dnf module enable php:remi-8.3 -y
56 sudo dnf install php php-cli php-fpm php-mysqld php-xml php-mbstring php-json php-zip -y
57 php -v
58 sudo vim /etc/php.ini
59 sudo dnf install php php-fpm -y
60 sudo systemctl enable --now httpd && sudo systemctl enable --now php-fpm
61 sudo vim /var/www/html/info.php
62 sudo firewall-cmd --permanent --add-port=80/tcp --add-port=443/tcp && sudo firewall-cmd -
-reload
63 history
```

```
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost bharath9399]# sudo systemctl enable --now httpd
[root@localhost bharath9399]# sudo systemctl enable --now php-fpm
Created symlink /etc/systemd/system/multi-user.target.wants/php-fpm.service → /usr/lib/systemd/system/p
hp-fpm.service.
[root@localhost bharath9399]# sudo vim /var/www/html/
[root@localhost bharath9399]# cd /var/www/html
[root@localhost html]# ls
batch8.html
[root@localhost html]# touch info.php
[root@localhost html]# ls
batch8.html info.php
[root@localhost html]# nano info.php
[root@localhost html]# sudo vim /var/www/html/info.php

[1]+  Stopped                  sudo vim /var/www/html/info.php
[root@localhost html]# sudo vim /var/www/html/info.php
[root@localhost html]# sudo vim /var/www/html/info.php
[root@localhost html]# ls
batch8.html info.php
[root@localhost html]# nano info.php
[root@localhost html]# sudo firewall-cmd --permanent --add-port=80/tcp
success
[root@localhost html]# sudo firewall-cmd --permanent --add-port=443/tcp
success
[root@localhost html]# sudo firewall-cmd --reload
success
[root@localhost html]#
```

PHP 8.3.24 - phpinfo()

← → 🔍 Not secure 192.168.163.134/info.php

PHP Version 8.3.24



System	Linux localhost.localdomain 5.14.0-596.el9.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Jun 27 16:02:33 UTC 2025 x86_64
Build Date	Jul 29 2025 15:48:33
Build System	Red Hat Enterprise Linux release 9.6 (Plow)
Build Provider	Remi's RPM repository <https://rpms.remirepo.net/> #StandWithUkraine
Compiler	gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-5)
Architecture	x86_64
Server API	FPM/FastCGI
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php.d
Additional .ini files parsed	/etc/php.d/10-opcache.ini, /etc/php.d/20-bz2.ini, /etc/php.d/20-calendar.ini, /etc/php.d/20-ctype.ini, /etc/php.d/20-curl.ini, /etc/php.d/20-dom.ini, /etc/php.d/20-exif.ini, /etc/php.d/20-fileinfo.ini, /etc/php.d/20-ftp.ini, /etc/php.d/20-gettext.ini, /etc/php.d/20-iconv.ini, /etc/php.d/20-mbstring.ini, /etc/php.d/20-mysqli.ini, /etc/php.d/20-pdo.ini, /etc/php.d/20-phar.ini, /etc/php.d/20-simplexml.ini, /etc/php.d/20-sodium.ini, /etc/php.d/20-sockets.ini, /etc/php.d/20-tokenizer.ini, /etc/php.d/20-xml.ini, /etc/php.d/20-xmlwriter.ini, /etc/php.d/20-xsl.ini, /etc/php.d/30-mysql.ini, /etc/php.d/30-pdo_mysql.ini, /etc/php.d/30-pdo_sqlite.ini, /etc/php.d/30-xmlreader.ini, /etc/php.d/30-zip.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no
Thread Safety	disabled
Zend Signal Handling	enabled

